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CS-2402

B. Tech. CSE (Third Semester)

EXAMINATION, 2023-24

DATA STRUCTURE USING 'C'

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any *five* questions. Each question carries 8 marks. $5 \times 8 = 40$

1. ✓ What is B Tree ?

2. ✓ Define link list and its type ✓

3. Explain Graph, and outline a directed graph with example. ✓

P. T. O.

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4. Write divide and conquer approach with example.
5. Define Array with example.
6. Write Algorithm for BFS Search.
7. Define ADT.
8. Differentiate b/w linear and non-linear data structure.
9. Write the condition to place a new node on a list.
10. Define Queue and its operation.
11. Explain binary search and linear search.
12. Define circular queue and also write pseudo code to add element in queue.
13. Describe B tree. Also explain benefits of ADT.
14. What is binary heap?
15. What is sorting? Sort a given array with the help of heap sort.
12, 3, 11, 3, 2, 7, 4

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Section—B

(Long Answer Type Questions)

Note : Attempt any two questions. Each question carries 30 marks. $2 \times 30 = 60$

1. Define complete binary tree? What is an almost complete binary tree? Also discuss AVL Tree.
2. (a) Explain hashing, hash table and a function. Explain with an example.
(b) Compare and analyse sequential search, binary search.
3. What is Stack? Write the applications of stack also discuss algorithm to implement push and pop operation of stack.
4. Explain selection sort and sort array 10, 05, 20, 15, 33, 6, 22, 11 and discuss its best, worst, average case complexity.
5. Define tree? Prove that a tree with n vertices exactly $n-1$ edges.

P. T. O.

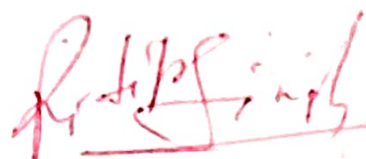
6. Write short note on the following :

~~(A)~~ Abstract Data Type

(B) Hash function

~~(C)~~ Array

~~(D)~~ Double ended queue.


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B. Tech. (CS) (Third Semester)

EXAMINATION, 2023-24

DISCRETE STRUCTURE

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any *five* questions. Each question carries 8 marks.

$5 \times 8 = 40$

- ✓ 1. Two finite sets have m and n elements. The total number of subsets of the first set is 56 more than the total number of subsets of the second set. Find the values of m and n .

P. T. O.

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2. Prove that :

$$(A \cap B)^c = A^c \cup B^c$$

3. Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$ Find $A \Delta B$ i. e. (A symmetric difference B).

4. Define floor function and ceiling function with example.

5. A survey shows that 73% of the Indians like apples, whereas 65% like oranges. What percentages of Indians like both apples and oranges?

6. Let :

$$A = \{1, 2, 3\},$$

$$B = \{p, q, r\}$$

$$\text{and } C = \{x, y, z\}$$

$$\text{and let } R = \{(1, p), (1, r), (2, q), (3, q)\}$$

$$\text{and } S = \{(p, y), (q, x), (r, z)\}$$

Compute $R \circ S$.

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7. Prove that for any set A which contains n elements, then total different asymmetric relation will be $(3)^{\frac{n^2-n}{2}}$.

8. Prove that the number of functions from $A \rightarrow A$ is less than number of relations from $A \rightarrow A$ i.e. $n^n < 2^{n^2}$.

9. Show that $n^3 + 2n$ is divisible by 3 for $n \geq 1$.

10. Prove that $(A, +)$ is a semi-group, where A be the set of all positive even integers and + be the ordinary addition operation.

11. Define the following :

(i) Monoid

(ii) Abelian group

12. Define the following :

(i) Bipartite graph

(ii) Regular graph

P. T. O.

$n-2$

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13. If a complete graph have 5 vertices, then find the total number of spanning tree possible. 2^{n-2}
14. What is the chromatic number of C_n , a cycle with n vertices?
15. What is the value of the following prefix expression?

$\uparrow - * 3 3 * 4 2 5$

Section—B

(Long Answer Type Questions)

Note : Attempt any two questions. Each question carries 30 marks. $2 \times 30 = 60$

1. (a) Prove by mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n .
- (b) Let G be a group with identity e . Show that if $a^2 = e$ for all $a \in G$, then G is abelian.
2. (a) Simplify Boolean function algebraically and draw circuit diagram :
- $$f(x_1, x_2, x_3) = (x_1 x_2 + x_3) \cdot (x_2 + x_3) + x_3$$

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- (b) Consider the partially ordered set $(\{2, 4, 6, 8\})$, where " $2/4$ " means 2 divides 4.

Show with reason that whether the following statements are true or false :

- (i) Every pair of elements in the poset has greatest lower bound. (T)
- (ii) Every pair of elements in poset has a least upper bound. (F)
- (iii) This poset is a lattice. (F)

3. (a) Consider the following conditional statement :

P : If the floods destroy my house or the fires destroy my house, then my insurance company will pay me.

Write the converse, inverse and contrapositive of the statement.

P. T. O.

[6]

(b) Is the statement tautology

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r) ?$$

4. Does there exist a simple graph with the following degree sequences. Show with reason. (Using Havel Hakimi theorem) :

(i) 7, 5, 6, 4, 4, 3, 2, 1

(ii) 6, 6, 6, 3, 6, 2, 2

(iii) 3, 3, 3, 2

(iv) 7, 6, 4, 6, 4, 3, 2, 2

5. (a) If G is a connected planar graph, then prove that it has vertex with degree 5 or less.

(b) Consider a simple graph G , where $|E(G)| = 30$ $|E(\bar{G})| = 36$, then find the value of $|V(G)|$.

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6. (a) The preorder traversal sequence of binary search tree is :

30, 20, 10, 15, 25, 23, 39, 35, 42

Find the postorder traversal of same tree.

(b) Find the generating function of the following series :

1, -1, 1, -1, 1, -1,

$$\frac{1}{1+x}$$

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CS-2404

B. Tech. (CS) (Third Semester)

EXAMINATION, 2023-24

COMPUTER ORGANIZATION

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any *five* questions. Each question carries 8 marks.

$5 \times 8 = 40$

1. Distinguish the two address instruction and Zero address instruction format.
2. Give a brief description about general purpose register.

P. T. O.

[2]

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3. Describe immediate addressing mode.
4. Differentiate between RISC & CISC.
5. Explain Auxiliary Memory. *Random Access Mem*
6. What is addressing mode ? Explain any three addressing mode. *+*
7. What is the purpose of DMA ?
8. Explain Flynn's classification. *SIMD*
9. Differentiate Vector, Masking, Non-Masking Interrupt.
10. Illustrate status register.
11. Explain IEEE format. Convert (354) in to 32 bit format.
12. Explain fully associative mapping.
13. Point out the control word in computer organization. *STMD*
14. Describe Register Stack.
15. Classify the microoperation in computer organization. *SIMD, MISD, MIMD*

[3]

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Section—B

(Long Answer Type Questions)

Note : Attempt any two questions. Each question carries 30 marks. $2 \times 30 = 60$

1. Explain Booth Algorithm. Apply algorithm to multiply $(-5 * 4)$.
2. Point out purpose of pipelining with all stages and efficiency.
3. Explain Direct Cache Mapping. Evaluate a direct mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find :
 - (a) Number of bits in tag
 - (b) Tag directory size
4. Point out the arbitration method. Describe Daisy Chaining Method and Parallel Arbitration. *Central Arbitration*

P. T. O.

5. Illustrate virtual memory.

6. Convert :

(a) $(104)_{10}$ to Octal

(b) $(3084)_{10}$ to hexa

(c) $(1010001010111)_2$ to octal

(d) $(11101011001101010)_2$ to Hexa

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CS-2403

B. Tech. (CS) (Third Semester)

EXAMINATION, 2020

COMPUTER BASED NUMERICAL

STATISTICAL TECHNIQUE

Time : Three Hours

Maximum Marks : 100

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any *ten* questions. Each question carries 4 marks. $10 \times 4 = 40$

1. Three approximate values of number $\frac{1}{3}$ are given as 0.30, 0.33 and 0.34. Which of these three is the best approximation ?

P. T. O.

[2]

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2. If:

$$r = 3h(h^6 - 2)$$

find the percentage error in r at $h = 1$, if the percentage error in h is 5.

3. Find a positive real root of $x - \cos x = 0$ by bisection method correct upto 4 decimal places between 0 and 1.

4. Find the real root of:

$$2x - \log_{10} x - 7$$

correct to four decimal places using iteration method.

5. Using the method of false position, find the root of equation:

$$x^6 - x^4 - x^3 - 1 = 0$$

upto four decimal places.

6. Find the real root of the equation $x = e^{-x}$ using the Newton-Raphson method.

[3]

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7. Show relation between operators (proof):

(i) $\Delta = EV = VE = \delta E^{\frac{1}{2}}$

(ii) $E = e^{hD}$

8. Find the missing value of the following data:

$f(x)$	x
7	1
X	2
13	3
21	4
37	5

9. Find the cubic polynomial which takes the following values:

x	$f(x)$
0	1
1	2
2	1
3	10

P. T. O.

[4]

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10. From the following table of half-yearly premium for policies maturing at different ages, estimate the premium for policy maturing at the age of 63 :

Age	Premium (in ₹)
45	114.84
50	96.16
55	83.32
60	74.48
65	68.48

11. Use Gauss' forward formula to find a polynomial of degree four which takes the following values of function $f(x)$:

x	$f(x)$
1	1
2	-1
3	1
4	-1
5	1

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12. $f(x)$ is the polynomial of degree four and given :
 $f(4) = 270$, $f(5) = 648$, $\Delta f(5) = 682$,
 $\Delta^3 f(u) = 132$

find the value of $f(5.8)$ using Gauss's backward formula.

13. Use Stirling's formula to find y_{35} , given
 $y_{20} = 512$, $y_{30} = 439$, $y_{40} = 346$ and
 $y_{50} = 243$.

14. Given $y_0, y_1, y_2, y_3, y_4, y_5$ (fifth differences constant), prove that :

$$y_{2\frac{1}{2}} = \frac{1}{2}c + \frac{25(c-b) + 3(a-c)}{256}$$

where $a = y_0 + y_5$,

$b = y_1 + y_4$,

$c = y_2 + y_3$.

15. Using Newton's divide and difference formula, calculate the value of $f(6)$ from the following table :

x	$f(x)$
1	1
2	5
7	5
8	4

P. T. O.

[6]

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Section—B

(Long Answer Type Questions)

Note : Attempt any *three* questions. Each question carries 20 marks. $3 \times 20 = 60$

1. A rod is rotating in a plane. The following table gives the angle θ (in radians) through which the rod has turned for various values of times t (in sec) :

t	θ
0	0
0.2	0.12
0.4	0.49
0.6	1.12
0.8	2.02
1.0	3.20
1.2	4.67

Calculate the angular velocity and angular acceleration of the rod at $t = 0.6$ sec.

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Evaluate :

$$\int_0^6 \frac{dx}{1+x^2}$$

by using :

- (i) Simpson's $\frac{3}{8}$ rule
(ii) Trapezoidal rule
(iii) Weddle's rule

3. Given :

$$\frac{dy}{dx} = \frac{(y-x)}{(y+x)}$$

with $y = 1$ for $x = 0$. Find y approximately for $x = 0.1$ by Euler's method.

4. Fit a straight line to the following data :

x	y
0	1
1	1.8
2	3.3
3	4.5
4	6.3

P. T. O.

5. The pressure of the gas corresponding to various volume V is measured, given by the following data :

$V \text{ (cm}^3\text{)}$	$P \text{ (kg cm}^2\text{)}$
50	64.7
60	51.3
70	40.5
90	25.9
100	78

Fit the data to the equation :

$$PV^\gamma = C$$

6. (a) Write algorithm of Stirling's formula.
 (b) Write a program in 'C' for Lagrange's method of interpolation.